KPPSC PMS Geography Paper-I — Q.3: Rocks, Sedimentary Classification & The Rock Cycle

KPPSC PMS Master Examination Notes — Paper-I (Physical & General Geography)

Target Examination	KPPSC PMS (Provincial Management Service) — Khyber Pakhtunkhwa
Paper & Subject	Geography Paper-I (Physical & General Geography — 100 Marks)
EXAM QUESTIONS LINKED:	• **2022 Q.2:** "Provide a detailed account of sedimentary rocks and their most workable classification based on origin and composition. (20 Marks)" • **2018 Q.3:** "Discuss the main types of rocks along with examples and describe the geological rock cycle. (20 Marks)" • **2013 Q.3:** "Define rocks and minerals. Discuss the circumstances under which sedimentary rocks are formed and classified. (20 Marks)" • **2008 Q.1:** "Describe the rock cycle. How can we identify igneous rocks on the basis of mineral contents? Explain types of sedimentary rocks. (20 Marks)"
Target Syllabus	Section 2: Landforms — Rocks: Origin, Formation & Classification; Rock Cycle
Marks & Allocation	20 Marks | Strict 35-Minute Descriptive Writing Limit
Core Theoretical Thesis	Rocks constitute solid mineral aggregates classified genetic-petrographically into Igneous, Sedimentary, and Metamorphic classes. Sedimentary rocks, representing 75% of continental surface exposures, evolve through weathering, transport, deposition, and diagenesis (compaction and cementation) into mechanically formed (clastic), chemically precipitated, and organically derived strata. The James Hutton Rock Cycle illustrates the continuous petrological transformation where endogenic tectonic heat and exogenic denudational processes cycle matter perpetually.

🎯 Exam Questions Linked

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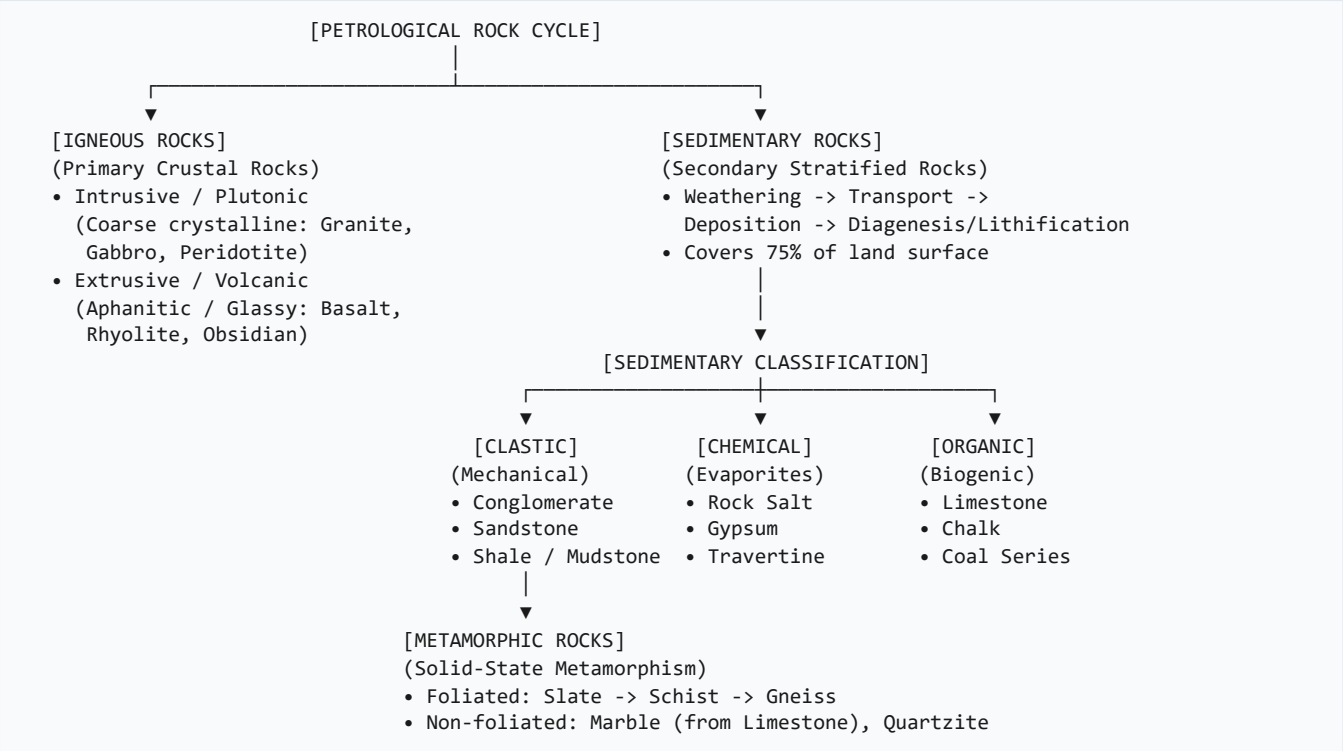
ROCK VS. MINERAL DISTINCTION: A mineral is a naturally occurring, inorganic crystalline solid with a definite chemical composition (e.g., Quartz SiO_2, Calcite $CaCO_3$). A rock is an aggregate of one or more minerals (e.g., Granite = Quartz + Feldspar + Mica; Limestone = Calcite).

THE THREE GENETIC ROCK FAMILIES: (1) Igneous (Primary rocks formed by magma crystallization; intrusive plutonic vs. extrusive volcanic); (2) Sedimentary (Secondary rocks formed via lithification of eroded detritus or chemical/organic precipitation; characterized by stratification, fossils, and porosity); (3) Metamorphic (Tertiary rocks formed by solid-state recrystallization under heat, pressure, and chemically active fluids).

DETAILED SEDIMENTARY CLASSIFICATION: Clastic/Mechanically formed (Conglomerate/Breccia > Sandstone > Siltstone > Shale based on Udden-Wentworth grain size scale); Chemically formed (Evaporites: Halite/Rock Salt, Gypsum; Carbonates: Travertine); Organically formed (Biogenic Carbonates: Chalk, Coral Limestone; Carbonaceous: Peat -> Lignite -> Bituminous Coal).

HUTTONIAN ROCK CYCLE: Formulated by James Hutton (1785) under the principle of Uniformitarianism ('The present is the key to the past'), demonstrating that rocks are not static relics but dynamic stages in a closed terrestrial matter-energy cycle.

1. Conceptual Mind Map & Thematic Roadmap



2. Analytical Dimensions & Geomorphic/Systemic Matrix

Sedimentary Rock Class	Sub-Types & Grain Size / Chemical Origin	Diagnostic Lithological Examples & Environments
Clastic (Mechanically Formed) — Rudaceous	Coarse gravels (>2 mm); rounded pebbles (conglomerate) vs. angular fragments (breccia).	Alluvial fans, high-energy mountain streams, scree slopes (e.g., Potohar Conglomerates, Salt Range Breccias).
Clastic (Mechanically Formed) — Arenaceous	Medium sand grains (0.0625 to 2 mm); dominated by resistant quartz minerals.	Sandstones (quartz arenite, arkose, greywacke); desert sand seas, beach deposits, deltaic distributor channels (e.g., Khewra Sandstone).
Clastic (Mechanically Formed) — Argillaceous	Fine particles (<0.0625 mm); silt (0.004–0.0625 mm) and clay (<0.004 mm); highly fissile bedding.	Shales, mudstones, claystones; low-energy environments: lake beds, floodplains, deep marine abyssal zones; primary source rock for petroleum.
Chemically Precipitated (Inorganic)	Precipitation of dissolved mineral salts due to evaporation, supersaturation, or pH shifts.	Evaporites: Rock Salt (Halite, NaCl) and Gypsum ($CaSO_4 \cdot 2H_2O$) formed in barred basins; Travertine/Tufa in Karst limestone caves and hot springs.
Organically Formed (Biogenic Carbonate)	Accumulation of skeletal remains, shells of marine organisms (calcite/aragonite $CaCO_3$).	Fossiliferous limestone, chalk (microscopic foraminifera), coquina, coral reef limestone (e.g., Margalla Limestone of Hazara/KPK).
Organically Formed (Carbonaceous)	Lithification of compressed plant/vegetal organic debris in anoxic swamp conditions.	Coal series: Peat (50% carbon) -> Lignite (60-70%, Thar coal) -> Bituminous (80%, Dandot) -> Anthracite (>90% metamorphic).

3. 20-Mark Master Model Essay

Introduction: Petrographic Foundations and the Earth's Crustal Architecture

Rocks constitute the solid mineral aggregates that compose the Earth's crust and lithosphere, forming the elemental substrate of all physical landscapes. In geological science, a rock is distinguished from a mineral: whereas a mineral is a naturally occurring, inorganic crystalline solid possessing a definite chemical formula (e.g., quartz SiO_2 or calcite $CaCO_3$), a rock is an aggregate of one or more minerals bound together by chemical crystallization or physical cementation.

Geoscientists classify rocks genetically into three great families based upon their mode of origin: Igneous rocks, formed through the cooling and solidification of molten magma or lava; Sedimentary rocks, produced through the denudation, deposition, and lithification of pre-existing materials; and Metamorphic rocks, formed through the solid-state recrystallization of existing rocks under extreme geothermal heat, directional pressure, and hydrothermal fluids. Although igneous and metamorphic rocks constitute 95 percent of the Earth's crust by volume, sedimentary rocks blanket over 75 percent of the continental surface, recording the environmental and biological history of the planet.

Genesis and Diagenesis of Sedimentary Rocks

Sedimentary rocks originate through an uninterrupted sequence of exogenic geomorphic processes operating across four sequential phases: weathering, transportation, deposition, and diagenesis (lithification). Mechanical and chemical weathering disintegrates exposed bedrock into detrital fragments and dissolved ionic solutions. Surface agents—predominantly fluvial systems, but also glacial ice, wind, and oceanic currents—transport these sediments down hydraulic gradients into sedimentary basins (lakes, river plains, continental shelves, and deep marine trenches). Once transport energy wanes, sediments settle into horizontal strata under gravitational influence (the Law of Original Horizontality). The conversion of unconsolidated sediments into solid rock occurs during diagenesis. As overburden accumulates, 'compaction' expels interstitial pore fluids and packs mineral grains tightly. Simultaneously, 'cementation' precipitates mineral-rich fluids (calcite, silica, iron oxides) across pore spaces, binding mineral grains into cohesive sedimentary strata. Distinctive features of sedimentary rocks include stratification (bedding planes), cross-bedding, ripple marks, porosity, and the unique preservation of organic fossils.

Comprehensive Classification of Sedimentary Rocks

Sedimentary rocks are categorized systematically into three principal genetic groups based on their petrological origin and composition: Mechanically Formed (Clastic), Chemically Precipitated, and Organically Formed (Biogenic) rocks. Clastic sedimentary rocks consist of weathered rock fragments (clasts) classified strictly according to grain diameter using the Udden-Wentworth scale. These subdivide into: Rudaceous rocks (coarse grains >2 mm), including rounded-clast Conglomerates deposited in high-energy braided rivers and angular-clast Breccias formed at the base of talus slopes; Arenaceous rocks (sand grains 0.0625–2 mm), dominated by durable quartz Sandstones found across desert ergs and beach shores; and Argillaceous rocks (fine silt and clay particles <0.0625 mm), forming fissile Shales and Mudstones in calm lacustrine and offshore marine environments. Shales constitute over 50 percent of all sedimentary rocks and serve as the paramount source and cap rocks in petroleum petroleum geology.

Chemical and Organic Sedimentary Strata: Carbonates and Evaporites

Chemically formed sedimentary rocks arise from the direct inorganic precipitation of dissolved mineral salts from supersaturated aqueous solutions, driven primarily by arid evaporation. Foremost among these are the 'Evaporites', exemplified by Rock Salt (Halite, NaCl), Gypsum ($CaSO_4 \cdot 2H_2O$), and Anhydrite, which crystallize in enclosed arid desert play lakes or restricted marine lagoons—such as the massive Neoproterozoic evaporite deposits of the Salt Range in Punjab/KPK. Travertine and calcareous tufa represent chemical carbonates precipitated around geothermal springs and within Karst caverns (stalactites/stalagmites). Conversely, Organically Formed (Biogenic) rocks derive from the accumulation of carbonaceous or calcareous organic debris. Biogenic carbonates include Fossiliferous Limestone, Coquina (cemented shell fragments), and Chalk (microscopic coccoliths and foraminifera shells). Carbonaceous rocks develop from the burial and biochemical coalification of dense plant matter in stagnant, anoxic swamps, progressing through the coal rank series: raw Peat (~50% carbon) to Lignite brown coal (~60–70% carbon, exemplified by the massive Thar coal deposits in Sindh), Bituminous coal (~80% carbon), and ultimately metamorphic Anthracite (>90% carbon).

The Huttonian Rock Cycle: Closed-System Terrestrial Matter Cycling

The relationship between igneous, sedimentary, and metamorphic rocks is synthesized through the petrological Rock Cycle, formulated in 1785 by Scottish geologist James Hutton under the doctrine of Uniformitarianism ('The present is the key to the past'). The rock cycle demonstrates that no rock type enjoys permanent existence; rather, Earth functions as a closed geochemical recycling system driven by two primary energy sources: exogenic solar radiation (powering climate, weathering, and erosion) and endogenic terrestrial radiogenic heat (powering plate tectonics and mantle melting). Primary igneous rocks (e.g., granitic plutons or basaltic plateaus) are elevated by tectonic orogeny and subjected to atmospheric weathering, yielding sediments that lithify into sedimentary rocks. As tectonic burial, subduction, or contact metamorphism exposes these sedimentary strata to elevated temperatures (>200°C) and

lithostatic pressures, they recrystallize into metamorphic rocks (e.g., shale metamorphosing into slate, schist, and gneiss; limestone recrystallizing into marble). If temperatures surpass melting thresholds (~700°C–1,200°C), metamorphic rocks melt completely into magma, which upon cooling crystallizes into new igneous rock, completing the perpetual cycle.

Conclusion: Lithological Dominance in Geomorphic Evolution

In evaluative summary, rocks and the geological rock cycle establish the fundamental lithological control over landform evolution. The differential resistance of rock types dictates topography: resistant sandstones and igneous granites form towering cliffs, cuestas, and mountain ridges, whereas soft, fissile shales and clays weather rapidly into broad alluvial strike valleys and badlands. In Pakistan's geomorphic landscape, the sedimentary sequence of the Salt Range, Potohar Plateau, and Sulaiman fold belts preserves a complete lithostratigraphic record spanning from the Precambrian salt marls to the Siwalik molasses. Understanding sedimentary classification and the dynamic rock cycle is indispensable not only for theoretical geomorphology, but also for economic resource extraction, including fossil fuel reservoirs, industrial mineral deposits, and civil engineering infrastructure.

4. Scholarly Perspectives & Geomorphic Axioms

Scholar / Geologist	Theoretical Contribution / Concept	Exact Quotable Academic Thesis / Axiom
James Hutton (1785)	Theory of the Earth & The Rock Cycle	"*We find no vestige of a beginning, no prospect of an end; rocks decay to form strata, strata rise into mountains, in one unbroken cycle of terrestrial renovation.*"
Charles Lyell (1830)	Principles of Geology (Uniformitarianism)	"*The forces which alter rocks and shape the earth today—weathering, sediment deposition, volcanic intrusion—have operated with uniform energy across all geological time.*"
F.J. Pettijohn (1957)	Sedimentary Rocks	"*Sedimentary rocks are the definitive historical archives of the planet; their textures, mineral assemblages, and bedding planes record the hydraulic environments and life forms of vanished worlds.*"
Norman L. Bowen (1928)	The Evolution of the Igneous Rocks	"*Fractional crystallization of a cooling basaltic magma systematically dictates the sequence of silicate minerals formed, from olivine to quartz, determining crustal rock compositions.*"
Arthur N. Strahler (1969)	Physical Geography	"*The structural form of any landscape is a direct reflection of underlying lithology; rock resistance to exogenic denudation remains the primary determinant of landform relief.*"